DIPLOMA PROGRAMME PRACTICE / SL / SET B

Economics / Paper 2

Question booklet · 105 minutes · 40 marks

Choose one of the two cases and answer all parts (a) to (g) of that case. A calculator is permitted. Use the extracts and data in your answers. Show workings and units, and draw fully labelled diagrams on separate answer paper.

QUESTION 1 / 40 MARKS

Solar power and grid access in India

Text A — Buying capacity or buying power. India's solar auctions and investment in transmission illustrate a problem of coordination. In the practice state, generators compete for long contracts. Once connected, solar generation can displace fossil power during daylight. Several substations cannot carry all midday generation. Output the network cannot take is curtailed, and evening demand still relies partly on other sources.

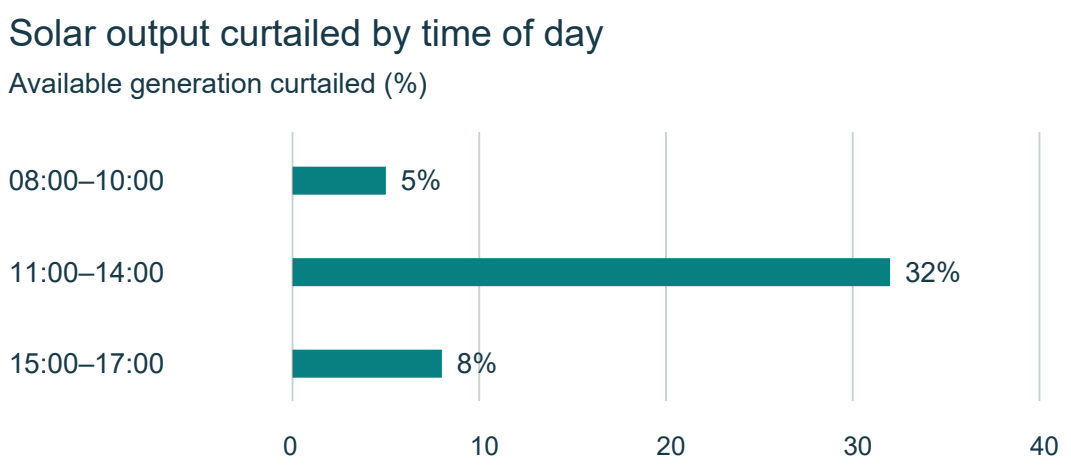
Text B — Two proposals. A production subsidy would pay for each megawatt-hour delivered to the grid. Alternatively, public investment would upgrade a congested substation and storage connection. Independent producers can enter the auction, but small firms report expensive credit. Network users include households receiving a subsidised retail tariff and industrial users facing more frequent interruptions.

Text C — A planning exercise. The proposed grid upgrade costs 600 million rupees. The exercise holds gross generation fixed, excludes operating costs and assumes each extra MWh delivered displaces generation worth 1,500 rupees. Forecasts are annual steady-state values, not official Indian statistics. No environmental benefit has been included in that saving.

Indicator	Existing grid	After upgrade
Gross solar generation (MWh)	500000	500000
Electricity curtailed (MWh)	100000	30000
Employment in installation during construction	0	450

Background reading: IEA, India Renewable Energy Market Report 2026. All district or scenario data and policy proposals are constructed for this paper.

FIGURE 1



Constructed district data for this case.

a(i) Define a subsidy in the context of the production proposal. **[2]**

a(ii) Define investment as a component of aggregate demand. **[2]**

b(i) Calculate electricity delivered before and after the grid upgrade, and the percentage increase relative to initial delivery. **[3]**

b(ii) Calculate the annual value of generation displaced by the additional delivery under Text C's assumptions. **[2]**

c Using a market diagram, explain how the production subsidy should affect the quantity of solar power delivered where the grid has spare capacity. **[4]**

d Explain why a subsidy for additional installed capacity could yield smaller environmental benefits than anticipated in Text A. **[4]**

e Using an AD–AS diagram, explain a possible long-run effect of more reliable grid infrastructure on industrial output. **[4]**

f Explain why 450 construction jobs cannot by themselves establish an increase of 450 in total employment. **[4]**

g Using the texts, data and your knowledge of economics, evaluate whether the state should prioritise grid investment over a production subsidy. **[15]**

QUESTION 2 / 40 MARKS

Tourism receipts and inclusion in Jamaica

Text A — Visitors and livelihoods. Jamaica's experience of a pandemic-related tourism interruption illustrates the risks of dependence on international travel. In the practice coastal district, reopened hotels generate foreign receipts. Some hotel groups import food under long contracts. Small local farms could supply more, but they lack cold storage and consistent certification. Hotel wages and job security differ across permanent and seasonal posts.

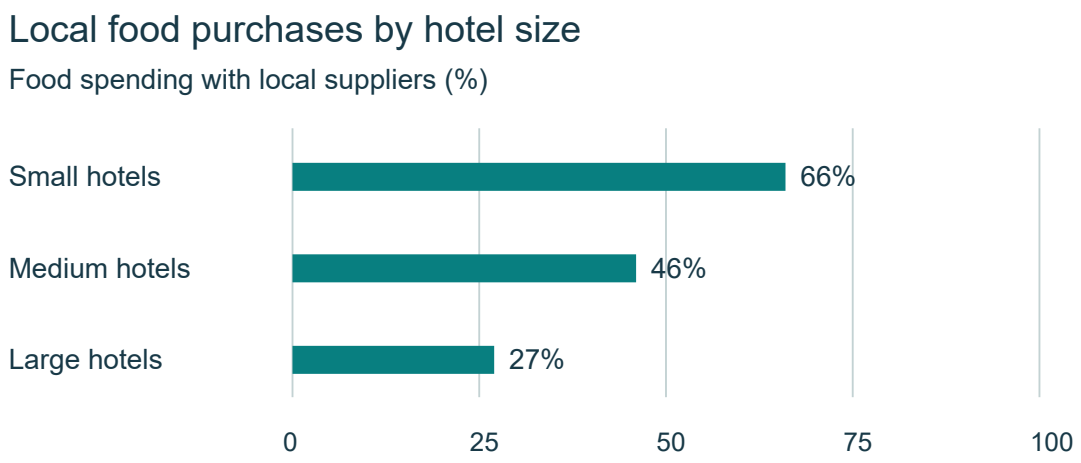
Text B — A development choice. The district could market additional visitor packages or co-finance cold storage and help farms meet buyers' standards. Marketing is quick to launch. More visitors could add pressure to a beach used by fishers and residents, while improved storage would also serve domestic markets. Land title requirements could exclude some small growers from the grant.

Text C — Interpreting receipts. The practice accounts cover only visitors' spending and imported inputs used directly to deliver tourist services. They exclude profit repatriation, subsequent rounds of spending and other industries. These are constructed district data, not national accounts.

District indicator	Year 1	Year 2
Visitor arrivals (thousands)	400	480
Spending per visitor (US dollars)	900	850
Imported inputs for visitor services (US dollar million)	108	144
Permanent local hotel jobs	7200	7350

Background reading: IMF, Jamaica Article IV consultation (2025). All district or scenario data and policy proposals are constructed for this paper.

FIGURE 1



Constructed district data for this case.

a(i) Define economic development. **[2]**

a(ii) Define a leakage from the circular flow of income and give one example from Text A. **[2]**

b(i) Calculate gross visitor receipts in each year and their percentage growth. **[3]**

b(ii) Calculate the change in visitor receipts after subtracting the imported inputs in the table. **[2]**

c Using a foreign exchange diagram, explain the likely effect of additional foreign visitor spending on the Jamaican dollar in a floating exchange market, other things equal. **[4]**

d Explain why increased demand for hotel meals might lead to only a small increase in local farm incomes in the short run. **[4]**

e Using a negative externality diagram, explain why the unregulated number of beach visits may exceed the socially efficient number. **[4]**

f Explain why the change in permanent hotel employment is insufficient evidence of a comparable improvement in household welfare. **[4]**

g Using the texts, data and your knowledge of economics, evaluate whether co-financing local food supply chains would support more inclusive development than additional tourism marketing. **[15]**